House Price Prediction Using Linear Regression

1. Objective

The objective of this project is to build and evaluate a Linear Regression model for predicting house prices using the California Housing dataset. The model learns the relationship between housing features and median house values, enabling price prediction for unseen data.

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2. Dataset Description

The California Housing dataset contains information collected from the 1990 California census. Each record represents a housing block group and includes various features affecting house prices.

Features

Feature	Description
MedInc	Median income in the block
HouseAge	Median house age
AveRooms	Average number of rooms
AveBedrms	Average number of bedrooms
Population	Population of the block
AveOccup	Average occupancy per household
Latitude	Geographic latitude
Longitude	Geographic longitude

Target Variable

- **MedHouseVal**: Median house value (in units of $100,000).

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset structure and characteristics.

Key Observations

- The dataset contains no missing values.

- Median income (MedInc) shows a strong positive relationship with house prices.
- Geographic features (Latitude and Longitude) influence housing values significantly.
- Some features contain outliers, especially Population and AveOccup.
- House prices are not perfectly linearly related to all features, indicating potential room for more advanced models.

Data Preparation

- Features and target variable were separated.
- The dataset was split into training (80%) and testing (20%) sets.
- No categorical variables were present, so encoding was unnecessary.
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4. Model Development

Algorithm Used

Linear Regression

Linear Regression models the relationship between independent variables and a dependent variable using a linear equation.

The model can be expressed as:

Price = $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$

where:

- β_0 = Intercept
- $\beta_1 \ldots \beta_n$ = Feature coefficients
- $X_1 \ldots X_n$ = Input features

Training Process

1. Load the California Housing dataset.
2. Split data into training and testing sets.
3. Train a Linear Regression model using training data.
4. Predict house prices on test data.
5. Evaluate model performance using standard metrics.

5. Model Evaluation

The following metrics were used to evaluate performance:

Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted values.

Mean Squared Error (MSE)

Measures the average squared prediction error.

Root Mean Squared Error (RMSE)

Provides prediction error in the original target units.

R^2 Score

Indicates how much variance in house prices is explained by the model.

Sample Results

Metric	Value
MAE	≈ 0.53
MSE	≈ 0.56
RMSE	≈ 0.75
R^2 Score	≈ 0.58

Interpretation

- The model explains approximately 58% of the variation in house prices.
- Prediction errors are moderate, indicating reasonable but not perfect performance.
- Linear Regression provides a useful baseline model for house price prediction.

6. Visualization

An Actual vs Predicted scatter plot was generated.

Observation

- Points close to the diagonal line indicate accurate predictions.
- Some deviation exists for high-priced houses, suggesting the model struggles with extreme values.

- Overall, the model captures the general trend of housing prices.

7. Improvement Ideas

Several improvements can enhance prediction accuracy:

1. **Feature Engineering**
    - Create new features from existing variables.
    - Include interaction terms.
2. **Feature Scaling**
    - Standardize numerical features before training.
3. **Outlier Handling**
    - Detect and remove extreme values.
4. **Cross-Validation**
    - Use k-fold cross-validation for more reliable evaluation.
5. **Advanced Models**
    - Random Forest Regressor
    - Gradient Boosting
    - XGBoost
    - Support Vector Regression
6. **Hyperparameter Tuning**
    - Optimize model parameters using Grid Search or Random Search.

8. Conclusion

A Linear Regression model was successfully developed for house price prediction using the California Housing dataset. The model achieved satisfactory performance and demonstrated the ability to predict housing prices based on socioeconomic and geographic factors. Although the model provides a strong baseline, further improvements through feature engineering and advanced machine learning algorithms can significantly enhance predictive accuracy.